

%nprocshared=64
%mem=320GB
%chk=PbRhaC10C106CT1.chk
opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc

PbRhaC10C106CT1

0 1

O	0.96862400	-0.55632200	-0.21341400
O	-0.02651600	-2.34228900	-2.37255700
O	-3.36841800	-2.75247100	-2.07963500
O	-0.58006000	0.77563200	-2.09047700
C	2.85786700	2.25509400	-2.63003000
O	3.22736900	2.10215400	-1.25869500
C	2.87230100	0.82990100	-0.77103100
C	1.35861700	0.69299500	-0.73326900
C	0.83891900	0.82415900	-2.13615000
C	1.32470300	2.12122200	-2.77709400
O	-2.69828000	-0.90303600	-3.07371200
O	-0.22724300	-3.44442600	-0.46782100
C	0.37165900	-3.23300300	-1.55031600
C	1.58063700	-4.07475500	-1.88602500
C	2.59239500	-4.13689600	-0.75297000
O	3.45144400	-2.96757100	-0.85354600
C	3.47209500	-5.36913800	-0.81343400
C	4.49795600	-5.44735700	0.30676200
C	5.32798700	-6.71999900	0.26601000
C	6.35351400	-6.78908300	1.38858700
C	7.19345400	-8.06208900	1.34868000
C	8.21948900	-8.14163500	2.47190000
C	9.05174700	-9.42058000	2.43291700
C	3.65391500	-2.21702200	0.23347800
O	3.02920000	-2.29473000	1.26028400
C	4.79374500	-1.25189800	-0.00740600
C	4.62126400	0.09056800	0.70346200
O	3.31588000	0.65244500	0.52644000
C	3.35679800	3.59320600	-3.09764500
O	0.89519200	2.21792000	-4.11187500
C	-3.92110300	0.88339800	3.73692900
C	-4.41824600	1.18109000	2.31259900
C	-3.27348600	1.73234100	1.46391000
O	-2.38733600	2.49271400	2.27189500
C	-1.61744500	1.64475100	3.13348700
C	-2.44896000	0.45544400	3.66398900
C	-0.33371300	1.20124500	2.45769700
O	-2.00940700	0.07850800	4.95518800
O	-3.78864100	2.53562600	0.45685500
O	-4.74783200	-0.07340400	4.35856300
O	-4.90798500	-0.03992400	1.78947700
C	-3.10890900	2.48920800	-0.80989400

C	-4.14242600	2.17610300	-1.86307400
C	-2.35426800	3.77413100	-1.10182500
C	-4.63725100	0.75179600	-1.76787500
O	-5.43167600	0.43497200	-2.76161300
O	-4.31425800	-0.01019900	-0.87450200
C	-5.91914300	-0.93724200	-2.95543500
C	-1.08110700	3.91615400	-0.27567700
C	-0.18417200	5.07145600	-0.71109700
C	1.17615200	5.02193300	-0.03644500
C	2.09247300	6.17329600	-0.44889000
C	3.51100700	6.03442300	0.08003500
C	4.42349700	7.16748500	-0.37436300
C	-4.84130400	-1.74514700	-3.69502700
C	-3.55088900	-1.83071700	-2.91469900
C	-6.46575000	-1.61295200	-1.70585700
C	-7.36353100	-0.73098600	-0.83681800
C	-7.65415600	-1.36723900	0.50736600
C	-8.24646400	-0.38876600	1.51883700
C	-8.39468300	-0.98434100	2.91031900
C	-8.84741000	0.03277800	3.95395200
C	-8.96526700	-0.56687600	5.34932500
C	5.72549600	1.06390400	0.30762700
C	5.68188500	2.36041100	1.09079900
C	6.71352800	3.37107500	0.62153500
C	6.77276500	4.61988300	1.49432200
C	7.77947500	5.65475800	1.01222000
C	7.83872100	6.89380800	1.88662000
C	8.81091000	7.94456700	1.35601700
H	3.33684200	1.44706900	-3.21367600
H	3.28982500	0.06222600	-1.44460300
H	0.96627900	1.51272500	-0.11127900
H	1.19827700	-0.03741000	-2.71864700
H	0.86429600	2.96273500	-2.25463900
H	1.20749200	-5.08058400	-2.08005000
H	2.05617400	-3.71521600	-2.80055900
H	2.06826700	-4.09170300	0.20706500
H	2.81402900	-6.25247200	-0.77637600
H	3.98048500	-5.39926900	-1.78995700
H	5.15709700	-4.57203300	0.26164300
H	3.97640300	-5.38398700	1.27512800
H	4.66463000	-7.59132200	0.31724400
H	5.84123200	-6.77601100	-0.70760900
H	7.01799600	-5.91841900	1.33390500
H	5.83884100	-6.72625700	2.35120700
H	6.53906300	-8.94300000	1.40369100
H	7.70841100	-8.13587200	0.38606400
H	8.88294100	-7.26140700	2.42478600
H	7.70318900	-8.06562900	3.44246200
H	9.78532300	-9.45345700	3.24226500
H	9.59975900	-9.49339600	1.48397900

H	8.41213500	-10.30272900	2.51998400
H	5.68662700	-1.72824400	0.38107400
H	4.95933300	-1.13702700	-1.07741100
H	3.10101700	3.75383900	-4.14587800
H	4.45107900	3.64598200	-2.99778400
H	2.91986600	4.39678000	-2.50390300
H	1.31489800	1.53640300	-4.63429000
H	-3.97574700	1.79260900	4.34874600
H	-5.22806100	1.92774300	2.34891000
H	-2.72272300	0.88551400	1.02397900
H	-1.38973900	2.27550500	4.00439000
H	-2.37744100	-0.40284100	2.98304900
H	0.27516900	0.58136500	3.13316000
H	0.27093500	2.07391100	2.17149600
H	-0.53639700	0.61313800	1.56382500
H	-1.18813700	-0.42394500	4.86909500
H	-5.01360600	-0.68844900	3.66414600
H	-4.79679200	-0.02692500	0.82043900
H	-2.38279700	1.67287900	-0.78132900
H	-5.00747000	2.84656100	-1.78318200
H	-3.73610100	2.31350100	-2.86607300
H	-2.07622800	3.75407200	-2.16946900
H	-3.01720400	4.63209800	-0.96648900
H	-1.33871700	4.02371600	0.78650300
H	-0.52527800	2.97842500	-0.34839700
H	-0.04826200	5.03873900	-1.79964700
H	-0.67968400	6.03167000	-0.49778900
H	1.05094800	5.03243000	1.05307400
H	1.67506600	4.07065900	-0.27456100
H	2.11752600	6.24662700	-1.54521700
H	1.66700700	7.12415500	-0.10439300
H	3.48777300	5.98844600	1.17581700
H	3.92455600	5.06737600	-0.25843700
H	5.43645800	7.05480000	0.01099300
H	4.04179400	8.13242300	-0.03721900
H	4.48613600	7.20612700	-1.47600000
H	-5.24696500	-2.75721900	-3.84870500
H	-4.64454200	-1.28433700	-4.66161900
H	-7.02871900	-2.48435000	-2.06293000
H	-5.63542600	-2.00385500	-1.10494100
H	-6.87577500	0.23226900	-0.66126500
H	-8.28895900	-0.50078200	-1.37204500
H	-8.32646100	-2.23856600	0.38735500
H	-6.72086700	-1.76716000	0.93140300
H	-7.58716100	0.48078100	1.56787200
H	-9.21141900	-0.01799900	1.15938200
H	-9.10193800	-1.82561600	2.88068800
H	-7.42772600	-1.40411000	3.22114000
H	-8.14114500	0.87392000	3.95841900
H	-9.80903000	0.45728200	3.64290000

H	-9.28185900	0.17633200	6.07655900
H	-7.99576600	-0.96616100	5.68140600
H	-9.68401800	-1.38356900	5.36692700
H	6.69290200	0.56112200	0.46135000
H	5.65079600	1.26667300	-0.76493300
H	4.68267900	2.79835400	1.01152900
H	5.83065500	2.14830000	2.15571100
H	7.70316800	2.89534800	0.59864500
H	6.49753100	3.65823200	-0.41347500
H	5.77739600	5.06751400	1.54733400
H	7.02155900	4.32107300	2.53133400
H	8.77731900	5.19903100	0.95869000
H	7.52254900	5.94185600	-0.02598900
H	6.83430900	7.32790800	1.95949200
H	8.11114800	6.60663700	2.90433300
H	8.83608000	8.82370000	2.00549100
H	8.52390700	8.27226000	0.35502800
H	9.82630200	7.54141000	1.29729700
H	-0.92551300	0.76686400	-2.98613800
H	1.47084300	-0.71613000	0.60444100
Pb	-1.65540600	-1.43689400	-0.91214300
H	4.65768700	-0.08905800	1.78100900
H	-6.73722800	-0.77125500	-3.65027600

C H O 0
6-311+g(2d,p)

Pb	0	
S	4 1.00	
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2 1.00	
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1 1.00	
	1.6600600927	1.0000000
S	1 1.00	
	0.80431655001	1.0000000
S	1 1.00	
	0.22627039020	1.0000000
S	1 1.00	
	0.84014530665D-01	1.0000000
P	3 1.00	
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3 1.00	
	2.2021426123	0.40196284806

	1.2209125119	0.46058131862
	0.63367559815	0.19367655397
P	1 1.00	
	0.28202837058	1.0000000
P	1 1.00	
	0.11333375666	1.0000000
P	1 1.00	
	0.43948707430D-01	1.0000000
D	6 1.00	
	61.315369628	0.33870800787D-03
	12.372195840	0.13788683942D-01
	6.9254944983	-0.75979608103D-01
	2.3319539939	0.28113784298
	1.2108730003	0.44474512269
	0.60090478506	0.35326874351
D	1 1.00	
	0.28135869813	1.0000000
D	1 1.00	
	0.11500000000	1.0000000
F	1 1.00	
	1.0000000	1.0000000
F	1 1.00	
	0.2896200	1.0000000

PB 0

PB-ECP 3 60

f potential

2

2	3.88751200	12.20989200
2	3.81196300	16.19029100

s-f potential

4

2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

p-f potential

6

2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

d-f potential

6

2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600

2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100